

Memoize

v1.4.1

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Memoize is a package for externalization of graphics and memoization of compilation results in general, allowing the author to reuse the results of compilation-intensive code. Memoize (i) induces very little overhead, as all externalized graphics is produced in a single compilation. It features (ii) automatic recompilation upon the change of code or user-adjustable context, and (iii) automatic externalization of TikZ pictures and Forest trees, easily extensible to other commands and environments. Furthermore, Memoize (iv) supports cross-referencing, TikZ overlays and Beamer, (v) works with all major engines and formats, and (vi) is adaptable to any workflow.

The two steps of externalization of graphics in Memoize

doc.tex

```
\documentclass{article}
\usepackage{memoize}
\usepackage{tikzlings}

\begin{document}

We all love Ti\emph{k}Zlings!
\tikz\penguin; is a penguin, \tikz\koala; is a koala, and \tikz\owl; is an owl.

\end{document}
```

doc.pdf

   We all love TikZlings!  is a penguin,  is a koala, and  is an owl.

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Using the externalized graphics

doc.pdf

We all love TikZlings!  is a penguin,  is a koala, and  is an owl.

This manual also documents packages Advice (v1.1.1) and CollArgs (v1.2.0). These are auxiliary packages which were developed alongside Memoize, but are distributed as independent packages as they might be useful outside the context of Memoize, as well. See sections 4.5.1 and 5.6.1 for Advice, and sections 4.5.2 and 5.6.3 for CollArgs.